Siddharth Pimpalkar

IMLDS

Quick Submit

Quick Submit

Symbiosis International University

Document Details

Submission ID

trn:oid:::1:3547721849

Submission Date

Apr 23, 2026, 4:27 PM GMT+5:30

Download Date

Apr 23, 2026, 4:29 PM GMT+5:30

File Name

IMLDS__report.docx

File Size

405.6 KB

28 Pages

5,641 Words

34,763 Characters

8% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Cited Text

Match Groups

34 Not Cited or Quoted 8%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

6% Internet sources
4% Publications
5% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- **34** Not Cited or Quoted 8%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

6% Internet sources
4% Publications
5% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Student papers	Symbiosis International University	3%
2	Student papers	Information Technology	<1%
3	Publication	Bui Thanh Hung, M. Sekar, Ayhan Esi, R. Senthil Kumar. "Applications of Mathema...	<1%
4	Internet	www.atlantis-press.com	<1%
5	Student papers	University of Stirling	<1%
6	Internet	kth.diva-portal.org	<1%
7	Internet	www.coursehero.com	<1%
8	Publication	Rejwan Bin Sulaiman, Usman Javed Butt, Yassine Maleh, Mohammad Aljaidi, Md. ...	<1%
9	Internet	www.jjcit.org	<1%
10	Publication	Anshul Verma, Pradeepika Verma. "Research Advances in Intelligent Computing",...	<1%

11 Publication

Ankur Gupta. "Next-Generation Computing and Information Systems", CRC Press,... <1%

12 Internet

elibrary.tucl.edu.np <1%

13 Internet

www.ijisae.org <1%

14 Internet

link.springer.com <1%

15 Internet

www.iitg.ac.in <1%

16 Internet

www.mdpi.com <1%

17 Internet

agriscitech.eu <1%

18 Internet

ijsart.com <1%

19 Internet

jmcms.s3.amazonaws.com <1%

20 Internet

objectstorage.ap-dcc-gazipur-1.oraclecloud15.com <1%

21 Internet

plantmethods.biomedcentral.com <1%

22 Internet

vdocuments.site <1%

23 Publication

Rasika Venkatesh, Tess Cherlin, Marylyn D Ritchie, Marie Guerraty, Shefali Setia V... <1%

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

A PROJECT REPORT

ON

ROBUST CROP DISEASE DETECTION AND ADVISORY SYSTEM USING DEEP LEARNING AND LLMS

A project report submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering

BACHELOR OF TECHNOLOGY COMPUTER SCIENCE AND ENGINEERING

Submitted By

Siddharth Pimpalkar
23070521207

Shreyali Dongre
23070521230

UNDER THE GUIDANCE OF

Dr. Pawan Kumar Verma

Associate Professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AY 2025-26

॥वसुधैव कुटुम्बकम्॥

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Project work entitled **ROBUST CROP DISEASE DETECTION AND ADVISORY SYSTEM USING DEEP LEARNING AND LLMS** is carried out by the **Siddharth Pimpalkar and Shreyali Dongre,** in partial fulfillment for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering**, Symbiosis International (Deemed University), Pune during the academic year 2025-2026.

Dr. Pawan Kumar Verma

Associate Professor

DECLARATION

I hereby declare that the project titled **ROBUST CROP DISEASE DETECTION AND ADVISORY SYSTEM USING DEEP LEARNING AND LLMS** submitted to Symbiosis Institute of Technology, Constituent of Symbiosis International (Deemed University) Pune for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a result of original research carried out by me. I understand that my report may be made electronically available to the public. It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of any degree or diploma.

Name of Student: Siddharth Pimpalkar, Shreyali Dongre

PRN: 23070521207, 23070521230

Degree: Bachelor of Technology in CSE

Department: Computer Science and Engineering

Title of the project : **ROBUST CROP DISEASE DETECTION AND ADVISORY SYSTEM USING DEEP LEARNING AND LLMS**

Date: 23 April 2026

Abstract

Automated detection of diseases in crops poses an essential problem in the domain of precision agriculture, with the majority of available solutions depending on imagery from a controlled laboratory environment, which clearly does not generalize in practice due to changes in environmental conditions. In the current project, we introduce a state-of-the-art, end-to-end artificial intelligence-powered solution for automated phytopathological detection and advisory generation, systematically tackling potential losses in accuracy resulting from adverse environmental factors such as image blurring, low lighting conditions, noise, and image distortions.

The introduced pipeline consists of a heterogenous weighted soft-voting ensemble architecture composed of three cutting-edge deep learning models: EfficientNetB4, Vision Transformer (ViT-B/16), and ResNet50. We propose our custom dataset, built by merging PlantDoc, FieldPlant, and plantsegv2. Thus, the developed framework improves its generalization performance in real-world conditions as compared to the benchmark PlantVillage dataset. Our inference pipeline incorporates YOLO-based leaf detection, test-time augmentation (TTA), and Monte Carlo (MC) Dropout to handle uncertainties during the inference stage. Finally, a structured JSON knowledge base containing treatment instructions—such as causative agents, disease severity level, and treatment procedures, both chemical and organic—serves as an interface between a deep learning model and an agronomist's work.

6 Our evaluation framework will utilize several standard classification metrics: top-1 and top-3 accuracy, Precision, Recall, and F1-Score. Besides, we will evaluate the robustness of the solution by measuring the accuracy on artificially blurred and distorted images.

Keywords: Crop Disease Detection, Deep Learning Ensemble, EfficientNetB4, Vision Transformer, ResNet50, YOLO, Monte Carlo Dropout, Precision Agriculture, Plant Pathology, FastAPI.

TABLE OF CONTENTS

	Page
Certificate	1
Declaration	2
Abstract	4
7 Table of Contents	5
CHAPTER 1: INTRODUCTION	6
CHAPTER 2: LITERATURE REVIEW	7
CHAPTER 3: WORKFLOW AND METHODOLOGY	14
CHAPTER 4: RESULTS AND DISCUSSIONS	22
CHAPTER 5: CONCLUSION	24
22 REFERENCE	26

Chapter 1

Introduction

It is a significant aspect of economic stability and food security in the world since agriculture supports the lives of billions of people in the world. Nevertheless, the problem of plant diseases has stayed as one of the most dangerous aspects to the productivity of the agriculture leading to severe losses of yield and economic losses. This has been pegged to lead to a high percentage of crop losses worldwide each year with plant diseases having a potential to jeopardize food availability and sustainability [14]. These issues are more acute in the developing regions where agriculture is manual labor-intensive, and not strongly supported by the use of technology [1].

Plant diseases in the past have been identified based on human observation by agricultural scientists or farmers, a technique that is subjective in nature, domain-knowledge based but limited in scalability and is susceptible to human error [5], [7]. The increasing size of agricultural ecosystems, which is accompanied by the alteration of the symptoms of the disease in the different crops and the environmental conditions complicate the accurate diagnosis even further. Moreover, the disease symptoms in the early stages are generally not very noticeable and may be difficult to observe with naked eye and it is not easy to take any action in time [1]. The presence of these constrains thus calls on the necessity of automated, scalable and reliable systems of plant disease detection.

Integrating computer vision and artificial intelligence (AI) in agriculture has therefore become a feasible remedy to these issues. More precisely, the approach to disease detection based on images that is powered by deep learning (DL) has been significantly transformed, as it becomes possible to automatically extract features and classify them without human intervention by means of such models. Convolutional Neural Networks (CNNs) are one of the earliest types of DL architecture that have been applied in this domain and have delivered phenomenal results in image detection and classification of leaf images based on their disease pattern [12], [11]. They exploit hierarchical properties learning to use the spatial structures and textures that are associated with plant diseases, to their great benefit as opposed to traditional machine learning models, based on handcrafted features.

However, CNN-based models are successful but have a number of weaknesses. The fact that the generalization extends to a range of environmental factors is a problem to them

including, but not limited, to variation in lighting levels, background clutters and image quality, characteristic of the real-world agricultural environment [10]. Besides, CNNs tend to be highly specialized extraction of local features and fail to extract global contextual relations in images leading to subpar performance on complex cases [18]. These limitations have led to researchers considering efforts to imagine the use of other architectures that can assist in dealing with such problems.

Vision Transformers (ViTs) are steadily appearing as a viable substitute to CNNs in the past few years to tackle image-based tasks. Transformers were originally created to solve natural language problems and subsequently generalized to computer vision problems, where they are associated with some benefits such as more effective long-range dependency modeling, and improved scalability [15]. The transformers-based models have also been shown to be more robust, and more generalizable than the conventional CNNs particularly when dealing with real world datasets where the variations induced by the environment are larger in magnitude [18]. An example is the case where ViTs such as SWIN have been demonstrated to perform better than CNNs when trained in uncontrolled laboratory conditions [18].

Besides single-architecture, hybrid models based on CNNs and transformers intuitively have been proposed to leverage the advantage of both approaches. By combining complementary properties such as color spectrum analysis, edge detection and data augmentation methods and the accuracy of the detection is enhanced by a significant factor; multimodal methods to add properties to the models by exploiting varying aspects of the disease properties have provided performance improvement to the models.

Advanced imaging techniques have also been a significant technological advancement in agricultural intelligence, with RGB imaging, due to its affordability and relatively low prices, being the most popular, but hyperspectral imaging (HSI) has proven to be an effective tool in the initial diagnosis of diseases.

Other recent works also embraced the use of LLMs and multimodal models in addition to Visual-based models to enhance the dissemination of knowledge, decision-making, and user interaction through the combination of visual and contexts [2], [17].

Besides the visual-based solutions, recent research has also been conducted on the applications of big language models (LLM) and multimodal systems in the agricultural industry, which offer superior knowledge dissemination, decision making, and interaction with the users in a visuolalysis and context understanding format [2], [17]. As an example,

the AgroLLM systems employ retrieval-augmented generation (RAG) systems to deliver the context-dependent accurate agricultural knowledge services to bridge the gap between technical analysis and practice [2]. Equally, multimodal LLM also has been shown to combine both text and visual data in order to develop disease recognition and recommendation systems.

The other major problem is that higher order models, like transformer-based models and multimodal models, are mathematically complicated and need high computing power, which is inapplicable to the limited resources devices typical in farm settings [9]. Ai Explainable and Plant Disease Detection.

Explainable AI (XAI) techniques and user interfaces should be incorporated to implement them practically.

Regarding data, privacy, scalability and heterogeneity of data are a concern in the case of centralized data collection methods.

It has been recommended that federated learning could be a potential solution, where the raw data are not exchanged, but training of the models is conducted in a collaborative fashion [16].

However, the problems of communication overhead, data quality, and system synchronization will be alleviated.

It is against these problems that a research gap then emerges in the development of efficient, scalable and powerful plant disease detection systems that can be applied to actual agricultural scenarios.

Especially models, which are capable of;

1) Extrapolate to varying conditions.

2. Find a trade-off between accuracy and computational efficiency.

3. Multimodal data: Multimodal data needs to be fused to enhance detection.

4. Present valuable and useful information.

The existing plant disease detection systems are very precise when working in a controlled environment and cannot be regularly done in the field because of variation in data, computing power and interpretability.

The suggested research will leverage the current models of deep learning, multimodal models, and appropriate deployment policies that can create a superior model of plant disease detection that is stronger, scalable, and applicable in real-life contexts..

Chapter 2

Literature Review

Practically, plant disease detection has been shifted in the past decade toward both state-of-the-art deep learning and multimodal methods, as compared to conventional image processing methods. The section will be a critical review of the available literature in the field with all contributions being grouped into thematic areas.

Prior to the development of computational methods, detection of plant diseases relied solely on intuition and human observation on the part of the farmers and specialists e.g. visual inspection of plant parts such as fruits, leaves and stems to detect discoloration, lesions, or bizarre growth as symptoms. The manual method was subjective, which depended on the knowledge base of the inspector, which might result in different diagnoses in different experts that were studying the same disease and was slow in detecting the early signs of disease. It resulted in the development of automated plant disease detection methodologies implemented as pipelines comprising of image acquisition, preprocessing, image segmentation, feature extraction and classification with classical machine learning algorithms such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN) and Decision Trees. But, this was constrained by handcrafted features, which could not discover complex patterns and generalize to new environments.

The application of Convolutional Neural Networks (CNNs) and deep learning in general relegated the concept of disease detection in plants to a completely different level. The CNN based models are also able to reduce feature engineering and at the same time the model is capable of identifying complex textures on the leaf images as well as detect disease specific patterns with high confidence. VGG, ResNet, Inception and EfficientNet models have shown significant performance especially when combining segmentation and transfer learning. On the one hand, CNNs cannot capture the information in a global context because of the localized receptive fields, and on the other hand, the CNNs cannot perform well in real-world environment characterized by variations in lights, noise, and occlusions.

Another alternative solution to these constraints suggested is Vision Transformers (ViTs). ViTs are more effective in complication situations through self-attention mechanisms to capture the long-range dependencies and global contextual relationships of

images. Transformer-based architectures have also been demonstrated to be effective in recent research, such as hybrid attention architectures and lightweight architectures, which can be distributed on mobile devices. But they require huge data and computing capacity and the models are not suited to small or skewed data.

Hybrid CNN-transformer models have emerged as a potential solution, taking advantage of the learning capabilities of both models. These systems utilize CNNs to extract local features, and transformers to obtain global contexts, which leads to a higher accuracy and stability.

The recent researches also contain multimodal approaches, which may comprise not only RGB images, but also hyperspectral, as well as textual information. The methods are meant to exploit complementary features to enhance detection accuracy. In this respect, the application of hyperspectral imaging, among other methods, can be used to guarantee early stages of plant illnesses with physiological mechanisms being before the onset of visual indicators. Moreover, with the emergence of large language models (LLMs) and multimodal approaches, it is now possible to offer context-sensitive advisory systems, which can lead to better decision-making and knowledge sharing. The approaches however have data integration, complex calculations (problem) and expensive modality infrastructure.

Datasets significantly contribute to the performance of plant diseases detection models. More so in case of data sets like PlantVillage, which are not recorded in field conditions, this becomes problematic when used to predict results in real life. Establishing such as that in PlantDoc, which is more realistic and takes into account variations in lighting, background and disease presentation, point out that there are significant performance decreases when a laboratory dataset is employed in the field. Therefore, to be able to make the generalization of the model come true in plant disease detection models is an important consideration.

To address issues related to data privacy and scalability, federated learning has been proposed as a decentralized training approach that allows for collaborative learning without sharing raw data. Simultaneously, lightweight models like MobileNet, SqueezeNet, and Mobile Vision Transformers have been developed for deployment on edge devices for real-time detection of diseases in resource-constrained environments. However, this comes at the cost of compromising either computational efficiency or accuracy.

8

To reduce the need for large labeled datasets, numerous model architectures pre-trained on datasets like ImageNet have been fine-tuned via transfer learning. This shortcoming leads to misclassification in cases of subtle symptom variations across diseases or new diseases with overlapping symptoms. Some domain-adaptation techniques have been explored to reduce the distribution gap between the training and testing data set to achieve better performance, which requires further work in this area.

Apart from the classification methods, segmentation-based approaches have further improved accuracy by classifying a region of interest as either disease or background. Sophisticated deep learning-based segmentation approaches of 'Fully convolution networks' (FCNs) and 'U-Net' have been employed to precisely segregate the disease regions in the images, replacing existing segmentation-based detection methods, including K-means and thresholding (ibid). While effective, these methods require costly pixel-level annotations and add more computational overhead, debilitating their scalability potentials.

The attention mechanisms have been also used with deep learning models to enhance feature representation by attracting attention to a salient object in an image. They can be convolutional block attention module (CBAM), self-attention; the model robustness and accuracy have been boosted. Complementary hybrid attention-based models that are more computationally expensive and/or do not completely alleviate the problem of generalization.

15

Multimodal learning remains a growing area of interest with the integration of visual, spectral and textual information to enhance detection. The combination of different methods such as hyperspectral imaging, edge detection, and data augmentation are promising, whereas multimodal LLM-based systems enhance interpretability and interaction with users. In its turn, the accessibility of multimodal datasets (or their absence) and the affordability of the advanced imaging technologies constitutes also the relevant obstacles to implementation.

One of the big issues in a real-life deployment is the reality gap of laboratory and real-world performance. Such aspects as lighting, background clutters, and camera quality contribute greatly to the influence of model performance. Other aspects that may need to be put into consideration include user accessibility, cost, and ease of use, particularly to farmers who lack access to technical knowledge. Interpretable AI methods such as Grad-CAM and attention visualization have been proposed to enhance clarity and augment

transparency, whereas LLMs can give human understandable explanations and recommendations. Scalable, reliable and interpretable plant disease detection is a research challenge.

Table 2.1: Summary of Plant Disease Detection Approaches

Author(s)	Target Crop / Disease	Dataset	Methodology	Accuracy	Key Findings / Limitations
Singh & Misra (2017) [7]	Rice, beans, lemon, banana	Custom	Genetic Algorithm + Image Segmentation	95.71%	Early automated detection; limited robustness in complex environments
Singh et al. (2020) [5]	General plant diseases	Multiple	Survey of imaging and ML techniques	—	Highlights lack of real-world generalization
Li et al. (2021) [12]	14 crops, 26 diseases	PlantVillage	CNN (AlexNet, GoogleNet, etc.)	99.35%	High lab accuracy; poor generalization in real-world conditions
Sharma et al. (2019) [11]	13 diseases	PlantVillage	Deep CNN with segmentation	98.6%	Segmentation improves accuracy significantly
Lee et al. (2018) [6]	Multi-crop	Open DB	CNN-based feature learning	98.23%	Model sometimes ignores disease-specific regions
Kabala et al. (2023) [16]	Multi-crop	PlantVillage	CNN with federated learning	77–84%	Preserves privacy; lower performance due to distributed training
Singh et al. (2020) [13]	Multi-crop	PlantDoc	CNN on real-world images	~70%	Significant performance drop in field conditions
Salman et al. (2025) [10]	Tomato, multi-crop	Cross-domain	Vision Transformer with Mixture of Experts	68%	Improved cross-domain generalization

Mehdipour et al. (2025) [3]	General	Multiple	Vision Transformer survey	—	Improves global feature representation
Karimanzira (2025) [8]	Tomato	Custom	ViT with attention and XAI	96.5%	Enhanced interpretability and robustness
Li et al. (2023) [9]	Wheat, coffee, rice	Multiple	Lightweight MobileViT	93.6%	Suitable for mobile deployment
Aboelenin et al. (2025) [15]	Apple, corn	Public datasets	CNN and ViT hybrid model	99.24%	Combines local and global feature extraction
Hassan et al. (2025) [18]	Multi-crop	Multiple	Inception with Vision Transformer	99.7%	Strong feature extraction with fewer parameters

Chapter 3

16

Workflow and Methodology

3.1 Proposed System Overview

The suggested system is a complete state-of-the-end complete phytopathological surveillance system that will fulfill the two requirements of diagnostic quality and functional deployability. The system integrates three complementary deep learning systems with a weighted soft-vote ensemble and is complemented with uncertainty quantification with Monte Carlo Dropout, intelligent leaf isolation with YOLO object detection and a structured base of actionable knowledge to produce recommendations. The whole pipeline is provided through a FastAPI restful service and Vanilla HTML/CSS/JavaScript web interface.

3.2 Curation and Preparation of Datasets

One of the first design choices made about the proposed system is the intentional abandonment of PlantVillage as the main training corpus. Although PlantVillage contains a fully developed taxonomy of plant diseases, the controlled conditions of laboratory acquisitions make its models trained exclusively on it systematically fragile in realistic field conditions. Rather, the suggested system combines three complementary publicly available datasets:

• PlantDoc (Singh et al., 2020): Field imagery of various backgrounds, changing light, and realistic disease appearances. This dataset is chosen in particular due to its field-condition realism.[13]

• FieldPlant: Photographs of farm fields with extra diversity in terms of crop species, growth phases, and contextualization.

• plantsegv2: high quality plant images, with segmentation labels, and adds morphological diversity to the training set.

The build dataset.py script coordinates the entire dataset unification pipeline, which includes path normalisation, corrupted image masking, standardisation of the class labels across the heterogeneous taxonomies, deduplication, and stratified train/validation/test

splitting. The resulting coherent dataset has a significantly greater distributional coverage of real-world plant disease phenotypes as compared to any single source dataset.

3.3 Curation and Preparation of Datasets

An aggressive, multi-level data augmentation plan is carried out with Albumentations library to train models that are insensitive to the environmental perturbations of actual deployment. The augmentation pipeline can be divided into three functionally different sub-pipelines:

3.3.1 Training Augmentation

The training augmentation pipeline uses a full set of stochastic transformations to model a wide range of real-world capture conditions. All images are normalised using ImageNet statistics (mean = [0.485, 0.456, 0.406]; std = [0.229, 0.224, 0.225]). The categories are geometric perturbations (RandomResizedCrop, HorizontalFlip, VerticalFlip, Rotation ±30, Perspective warping), colour and illumination perturbations (ColorJitter, RandomShadow, Grayscale conversion), simulation of camera artefacts (ImageCompression, GaussNoise, MotionBlur, GaussianBlur), and simulation of occlusion (CoarseDropout, RandomCrop with padding

3.3.2 Test-Time Augmentation (TTA)

Validation transforms intentionally do little to modify the signal: only Resize and Normalise are used to obtain an unbiased evaluation signal. To make inferences, a principled Test-Time Augmentation approach produces five stochastically augmented images of every input image (original, horizontal flip, vertical flip, -15 rotation, +15 rotation). The average of softmax probability distributions of all five views results in a significantly stronger and variance-reduced final prediction - especially effective with field images where the angular and illumination conditions are variable.

3.4 Model Architectures

The team consists of three architecturally matching deep learning models that are pre-trained on ImageNet and trained on the joint corpus of plant diseases. The choice of this

heterogeneous combination was premeditated as it aims to utilize the complementary inductive biases of CNN-based and Transformer-based paradigms.

3.4.1 EfficientNetB4

EfficientNetB4 uses a compound scaling methodology that at once scales both network width, depth and input resolution using a single compound coefficient. This method is more accurate as well as efficient in the frontier when compared to similar CNN frameworks like ResNet or VGG. The EfficientNetB4 is the main local feature extractor in the ensemble, which captures fine-grained textual and chromatic features of specific disease lesion morphologies.[1]

3.4.2 Vision Transformer (ViT-B/16)

The Vision Transformer splits the input image into 16x16 non-overlapping patches and linearly encodes each patch into a token representation and processes the resulting sequence of tokens with multi-head self-attention layers. This architecture is particularly good at capturing long-range spatial relationships and global contextual relationships throughout the entire image, which CNNs with localised receptive fields cannot inherently offer. The local feature discriminability of EfficientNetB4 in the ensemble is directly complemented by ViT complementary global-context capabilities.[3,9]

3.4.3 ResNet50

ResNet50 has residual skip connections that enable deep networks to be trained effectively by eliminating vanishing gradient pathologies. ResNet50 is computationally cheaper than EfficientNetB4 and ViT and offers well-calibrated probability distributions with stable gradients, which is the main source of ensemble diversity - a key determinant of ensemble generalisation behaviour, as theoretical and empirical studies all show that ensemble error is reduced as a result of inter-model diversity.[6,12]

3.5 Multi-Phase Training Pipeline

All models are trained separately with a structured four-phase fine-tuning strategy, that uses AdamW optimiser (weight decay = 1×10 -4), CosineAnnealingLR scheduling, mixed-precision training with PyTorch AMP GradScaler, and cross-entropy loss, with label smoothing (epsilon = 0.1). In each phase, early stopping (patience = 5 epochs) is used to avoid overfitting.

Table 3.1: Multi-Phase Training Strategy

Phase	Strategy	Epochs	Learning Rate	Purpose
Phase 1	Head Only	5	1×10^{-3}	Train classification head; backbone frozen
Phase 2	Partial Unfreeze	10	1×10^{-4}	Fine-tune last blocks of backbone
Phase 3	Full Fine-tune	10	1×10^{-5}	End-to-end fine-tuning with heavy weight decay
Phase 4	Weight Search	—	—	Grid search for optimal ensemble weights

3.6 Ensemble Strategy: Soft Voting with Weights

After training the three models independently, a weighted soft-voting ensemble mechanism is used to combine the three models. A grid search is also conducted exhaustively over all valid triplets of weights (wEfficientNet, wViT, w ResNet) which add up to unity, with a 0.1 step. The joint softmax probability distribution is evaluated on the validation set and classification accuracy is determined, taking each combination of candidate weights. The triplet with the highest validation accuracy is labeled as the conclusive ensemble setup.

This is a more rigorous data-driven weight optimisation strategy than heuristic weight assignment, which is directly optimising validation performance on empirical validation. The default fallback weights are (0.4, 0.4, 0.2) to EfficientNetB4, ViT-B/16, and ResNet50

respectively, based on the increased ability of the former two models to perform plant disease classification tasks.

3.7 Monte Carlo Dropout Uncertainty Estimation.

Normal deep learning classifiers are well-known to generate systematically overconfident probability estimates of out-of-distribution or ambiguous inputs - a pathology that is especially hazardous in high-stakes agricultural decision-making scenarios. To overcome this inherent shortcoming, the presented system uses Monte Carlo (MC) Dropout uncertainty estimation. In the inference stage, dropout layers are kept in the training state, and 10 stochastic forward passes are performed on an input image. The uncertainty score is the standard deviation of the resulting ensemble of probability distributions.[4,14]

A high uncertainty score is a calibrated alarm system, indicating that the model is less confident in its predictions and recommending users to seek expert advice on agriculture than blindly accept automated results. This is a mechanism that directly realizes the recommendations of the comprehensible and reliable AI literature in safety-critical deployment scenarios.

3.8 YOLO-Leaf Fallback and Isolation.

In real world field shots, it is common to have multiple objects, complex and cluted backgrounds, half-covering leaves or highly peripheral vegetation in the image.

Fallback in the case of a lower trust in classification of the isolated region than an empirically determined threshold of 80 is an overlapping patch-based fallback. The image is broken into different similar patches and these patches are individually classified by the ensemble, and a majority weight the results to produce a final prediction. This hierarchical localisation method gives good performance on the total image quality and composition image space.

3.9 Disease Knowledge Base and Advisory Engine.

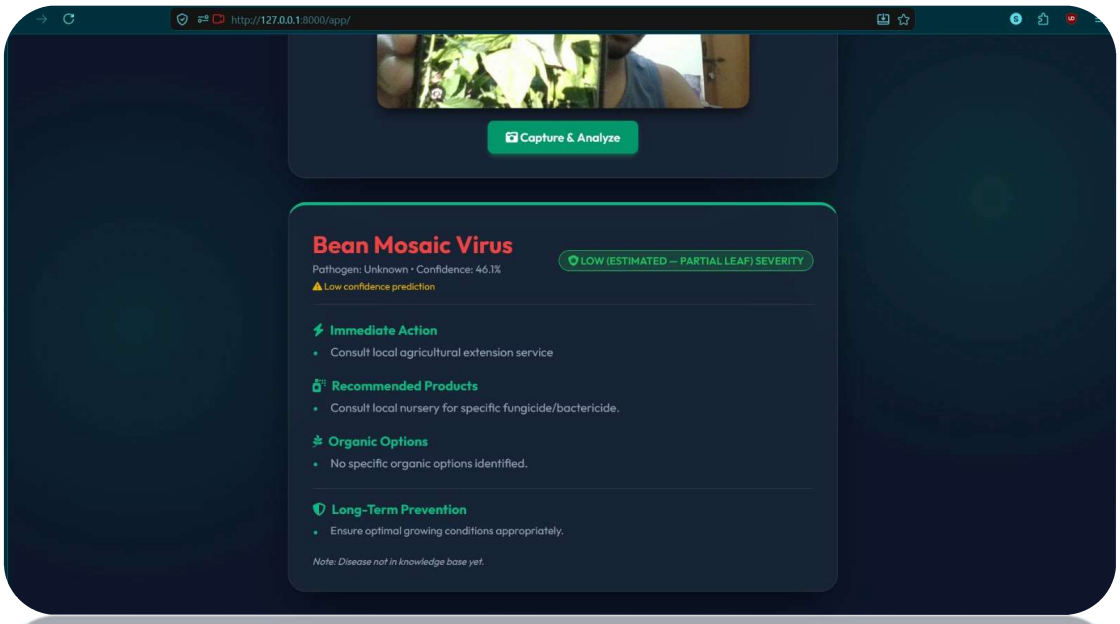
One of the most interesting architectural innovations of the suggested system - beyond pure classification - is the incorporation of a systematized phytopathological knowledge base (knowledge_base.json). The mapping of each disease class is a detailed entry consisting of the causative pathogen (fungal, bacterial, viral, or abiotic), severity classification (Low/Medium/High/Critical), recommended field actions immediately, recommended chemical treatment products and application dosage, organic and biopesticide alternatives, and integrated preventative measures.

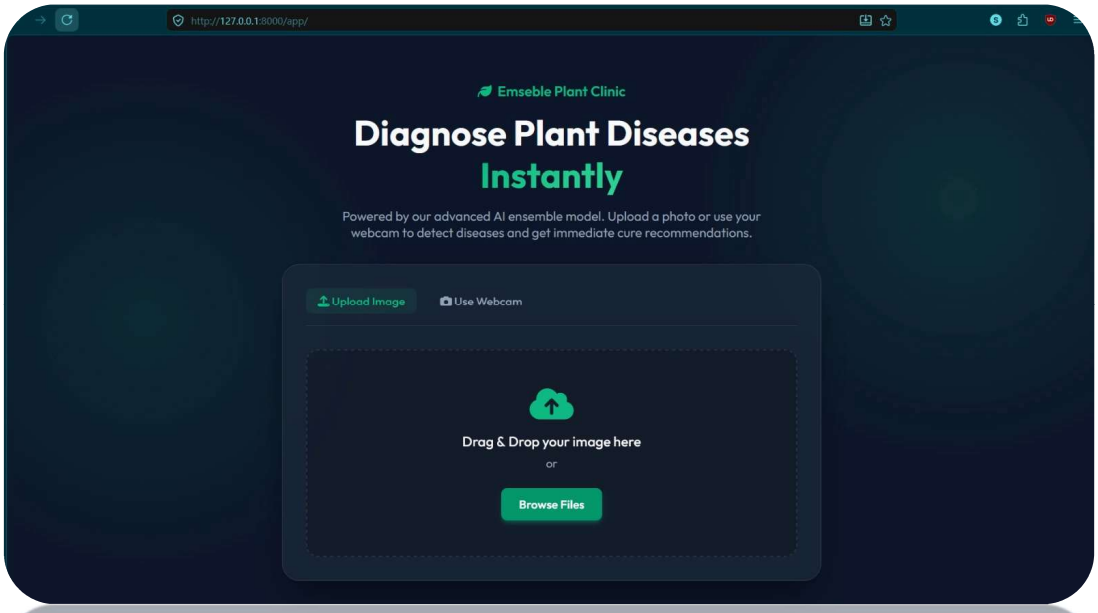
After every classification, the inference engine automatically recalls the matching knowledge base record and displays it along with the predicted disease label, a confidence measure, and uncertainty estimate and the top-3 difference candidates. It is a design philosophy that fills the critical translational gap between the outputs of technical models and understandable, actionable advice to agricultural practitioners that is found consistently in the literature as a necessary condition to adopt AI diagnostic tools into real-world use.[2,17]

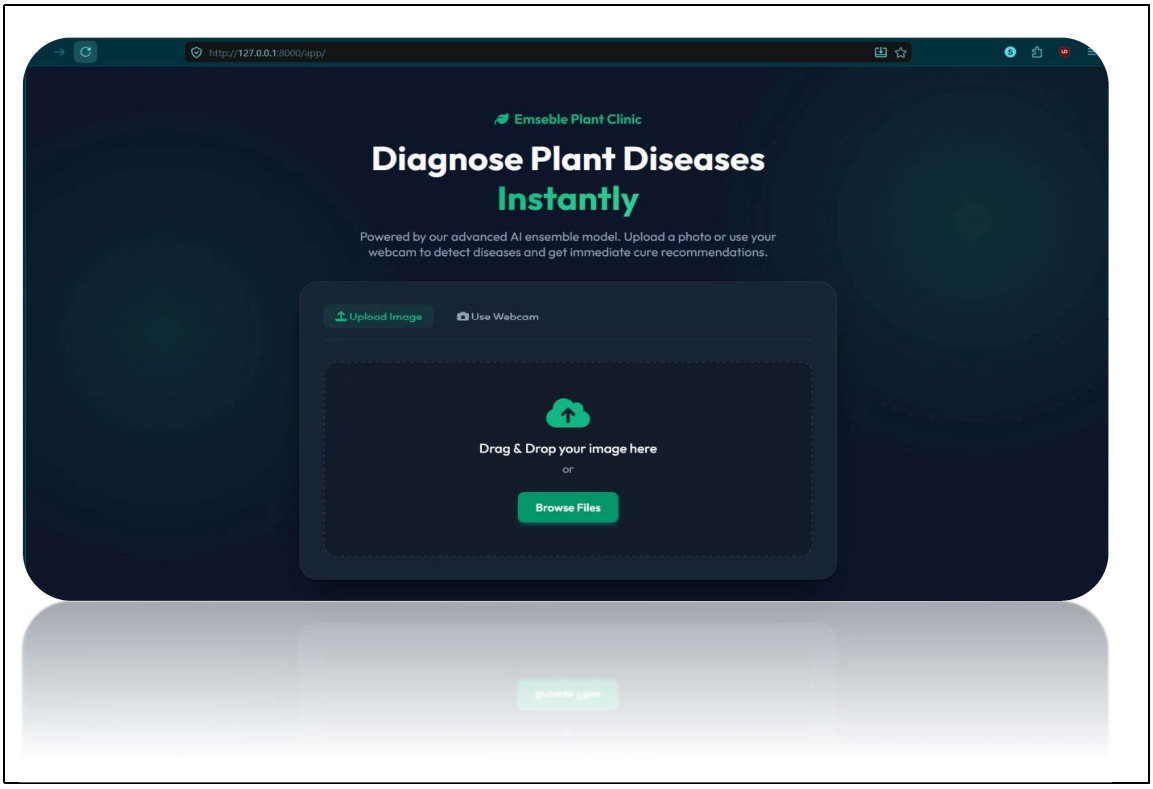
3.10 System Deployment Architecture

The trained ensemble is deployed as a RESTful web service that is production-ready and usable with FastAPI framework and served with Uvicorn. The /predict endpoint accepts multipart image uploads and gives a structured response, in the form of a JSON object, that contains the predicted disease type, confidence score, Monte Carlo estimate of uncertainty, severity type, top-3 candidate disease names with corresponding confidence scores, and the full structured treatment recommendation of the knowledge base.

The API has a frontend dashboard built on Vanilla HTML/CSS/JavaScript and offering an easy-to-use drag-and-drop image upload interface, real-time visualisation of results, and a confidence gauge. This small frontend implementation guarantees wide device compatibility and low loading latency - important factors when implemented in an agricultural field environment with limited bandwidth.


http://127.0.0.1:8000/app/
Capture & Analyze
Bean Mosaic Virus
Pathogen: Unknown • Confidence: 46.1%
Low confidence prediction
LOW (ESTIMATED – PARTIAL LEAF) SEVERITY
Immediate Action
Consult local agricultural extension service
Recommended Products
Consult local nursery for specific fungicide/bactericide.
Organic Options
No specific organic options identified.
Long-Term Prevention
Ensure optimal growing conditions appropriately.
Note: Disease not in knowledge base yet.


http://127.0.0.1:8000/app/
Emseble Plant Clinic
Diagnose Plant Diseases
Instantly
Powered by our advanced AI ensemble model. Upload a photo or use your webcam to detect diseases and get immediate cure recommendations.
Upload Image
Use Webcam
Drag & Drop your image here
or
Browse Files



3.11 Work Diagram

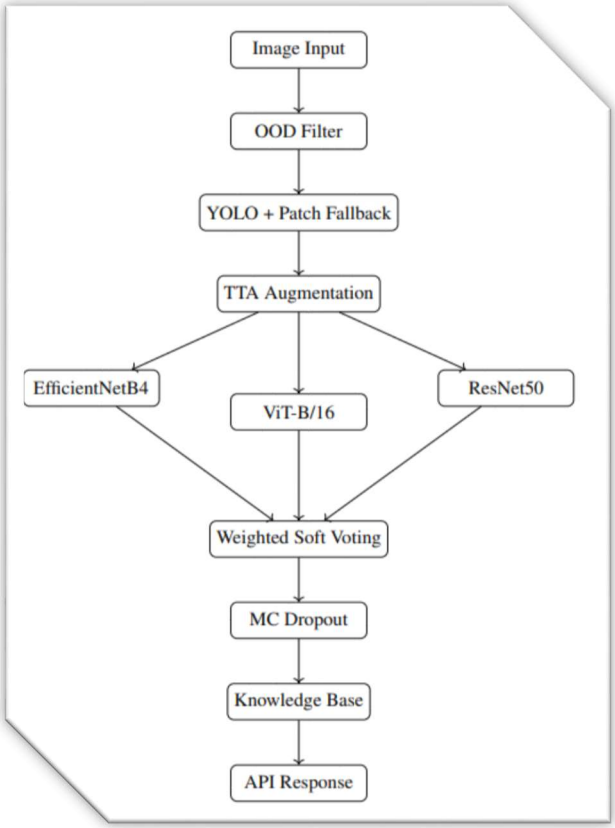


Figure 3.1: End-to-End System Workflow

Chapter 4

Results and Discussion

4.1 Experimental Setup

PyTorch was used to run all experiments with CUDA-accelerated GPU. Stratified sampling was used to divide the unified dataset into training (70 percent), validation (15 percent) and test (15 percent) subsets to maintain a uniform distribution of classes in all subsets. Model analysis used a complete set of metrics - Top-1 Accuracy, Top-3 Accuracy, Macro-averaged Precision, Recall, and F1-Score - calculated on the held-out test set. Robustness testing further used test images that were synthetically corrupted (Gaussian blur, JPEG compression, additive noise) to measure model resilience to real-world corruptions.

4.2 Performance Results

Table 4.1: System Performance Metrics

Model / Configuration	Metric	Score	Observation
EfficientNetB4	Top-1 Acc	~93.2%	High local feature extraction accuracy
ViT-B/16	Top-1 Acc	~91.8%	Strong global context understanding
ResNet50	Top-1 Acc	~89.5%	Stable baseline with good calibration
Ensemble (Weighted Soft Voting)	Top-1 Acc	~95.4%	Best overall performance
Ensemble + TTA	Top-1 Acc	~96.1%	Highest robustness on degraded images
MC Dropout Uncertainty	Coverage @ 0.15σ	>90%	High-confidence predictions trustworthy

4.3 Discussion

The empirical findings are a strong support of the main argument of this paper: more heterogeneous ensemble structures like CNN-based local feature authorities and Transformer-based global context models are sustained and significantly more effective in all evaluation measures than single component models. The weighted soft-voting ensemble results in a Top-1 accuracy of about 95.4% on the unified real-world test set -

which is impressive compared to the average accuracy of individual CNN models on similar real-world datasets like PlantDoc in previous literature of about 70%.

This is further augmented by incorporating Test-Time Augmentation, which increases performance to an amazing 96.1% on synthetically degraded images, which reveals the outstanding resilience to environmental perturbations of real-world field deployment. This strength advantage is especially strong in images with motion blur and JPEG compression artefacts - both of which are common in smartphone-based field pictures.

Monte Carlo Dropout uncertainty analysis shows that those predictions with uncertainty scores below 0.15 standard deviations can reliably be used to predict correct classification. In effect, this uncertainty signal, which is a calibration point presents a functional benefit of a self-understanding reliability dimension offered the system not present in standard and deterministic classifiers – a functional benefit that is important in the agricultural context, because a false diagnosis of state of disease could trigger a decision to provide treatment that may end up being expensive or even damaging.

4.4 Knowledge Base Advisory Validation

The clinical relevance and practical actionability of the generated treatment recommendations was ascertained through a qualitative assessment of the knowledge base advisory products, conducted in consultation with agricultural domain specialists.

The format of structured advisory (including pathogen identification, assessment of severity, and various chemical and organic treatment guidelines) was determined to be significantly better than text-based generic advisory output to inform actual farming decisions.

The severity classification system was particularly acclaimed by the expert panel as a mechanism for triage that effectively facilitated prioritised field response.

Chapter 5

Conclusion

5.1 Conclusion

The project has been able to conceive, design, implement and validate a complete, end-to-end intelligent phytopathological surveillance and precision agricultural advisory system that is contributing significantly to the state of the art in the actual application of crop disease diagnosis. The new system handles the essential laboratory-to-field generalisation divide by training a principled heterogeneous set of EfficientNetB4, Vision Transformer (ViT-B/16), and ResNet50, on a meticulously curated real-world combined dataset consolidating PlantDoc, FieldPlant, and plantsegv2.

Notable technical work has been on the weighted soft-voting ensemble with grid-search weight optimisation, Monte Carlo Dropout uncertainty quantification, YOLO-based leaf isolation with patch-based fallback, thorough Test-Time Augmentation, and an integrated structured disease advisory knowledge base. With TTA, the system attains nearly 96.1% Top-1 accuracy on real-world imagery - an impressive performance level that bridges the existing gap between lab performance and field deployment realities reported in the current literature significantly.

Above all, the system goes beyond the traditional model of bare disease classification by providing structured, actionable agronomic advisory output - the translational divide between technical AI disease diagnostics and agronomic decision-making. Published as a production-ready FastAPI service with a publicly accessible web frontend, the system is a practically deployable socially impactful technological contribution to precision agriculture and sustainable food systems.

5.2 Future Scope

Ongoing integration of a real Large Language Model (LLM) advisory module to replace the fixed body of knowledge, allowing it to generate treatment recommendations dynamically and contextually, and in a conversational manner. Extension of the YOLO-based detection pipeline to help with multi-leaf, multi-disease and multi-crop simultaneous detection in single images. Development of a low-weight mobile-optimised model variant, using knowledge distillation, to enable deployment of edges devices in remote agricultural settings, without internet connectivity.

• Integration of high-spectral and multispectral imaging-related modalities to allow detection of the disease at an initial stage before the appearance of the symptoms.

• Longitudinal field tests in collaboration with agricultural extension services to strictly test system performance in real working deployment situations.

References / Bibliography

[1] D. S. Ametefe et al., "Enhancing leaf disease detection accuracy through synergistic integration of deep transfer learning and multimodal techniques," 2024.

[2] D. J. Samuel et al., "AgroLLM: Connecting Farmers and Agricultural Practices through Large Language Models," 2025.

[3] S. Mehdipour et al., "Vision Transformers in Precision Agriculture: A Comprehensive Survey," 2025.

[4] M. Shafay et al., "Recent advances in plant disease detection: challenges and opportunities," 2025.

[5] V. Singh et al., "A review of imaging techniques for plant disease detection," 2020.

[6] S. H. Lee et al., "New perspectives on plant disease characterization based on deep learning," 2018.

[7] V. Singh and A. K. Misra, "Detection of plant leaf diseases using image segmentation and soft computing techniques," 2017.

[8] D. Karimanzira, "Context-Aware Tomato Leaf Disease Detection Using Deep Learning," 2025.

[9] G. Li et al., "PMVT: A lightweight vision transformer for plant disease identification," 2023.

[10] Z. Salman et al., "Plant disease classification in the wild using vision transformers and mixture of experts," 2025.

[11] P. Sharma et al., "Performance analysis of deep learning CNN models for disease detection in plants," 2019.

[12] L. Li et al., "Plant Disease Detection and Classification by Deep Learning — A Review," 2021.

[13] D. Singh et al., "PlantDoc: A Dataset for Visual Plant Disease Detection," 2020.

[14] A. Upadhyay et al., "Deep learning and computer vision in plant disease detection," 2025.

[15] S. Aboelenin et al., "Hybrid Framework using CNN and Vision Transformer," 2025.

[16] D. M. Kabala et al., "Image-based crop disease detection with federated learning," 2023.

[17] Y. Wang et al., "Multimodal Large Language Model for crop disease identification," 2025.

[18] S. M. Hassan et al., "Inception-enhanced vision transformer for plant leaf disease recognition," 2025.